Asya Azkin

asyaazkin12@gmail.com | 678-230-8003 | linkedin.com/in/asya-azkin | asya99.github.io/asyaazkin

SKILLS SUMMARY

Data visualization journalist and machine learning engineer specializing in turning complex public-interest data into clear, engaging, and accessible visual stories. **Experienced in analyzing large civic and government datasets and translating technical findings into charts, maps, and interactive narratives that inform broad audiences**. Actively bringing both rapid-turnaround visual reporting and deeper investigative data projects from analysis through storytelling.

Skills: Python, SQL, C#/.NET | Data Analysis, Machine Learning, NLP, Unsupervised Learning | Pandas, NumPy, scikit-learn, TensorFlow, PyTorch | UMAP, HDBSCAN | Data Visualization: Tableau, Power BI, Datawrapper, Flourish, Adobe Creative Suite, D3.js | Web & Publishing: WordPress, REST APIs | GIS & Spatial Thinking | AWS, Azure, Docker | Responsible AI, Investigative & Civic Data Research

EDUCATION

Georgia Institute of Technology | M.S in Computer Science | Specialization in Machine Learning |

Vanderbilt University | B.S. in Computer Science, Psychology

EXPERIENCE & LEADERSHIP

A.I. Engineer & Researcher | Emory University | November 2025 - Present

- Built a data analysis and visualization pipeline that transformed **~20M words of civic and political text into interactive visual maps**, enabling journalists and researchers to explore themes, narratives, and relationships across large document collections.
- Analyzed ~8,000 text segments using semantic embeddings and dimensionality reduction (UMAP) to model conceptual similarity and surface meaningful story patterns within complex public-interest datasets.
- Identified and validated thematic structure using clustering (HDBSCAN) combined with human review, ensuring findings were interpretable, accurate, and suitable for public-facing research and storytelling.
- **Designed a modular workflow spanning data acquisition, cleaning, analysis, and visualization**, and collaborated with interdisciplinary teams to integrate results into an accessible web application.

Software Engineer 1 | Cox Enterprises | April 2024 - July 2025

- Built and deployed data pipelines and analytics dashboards tracking performance across automation workflows, processing **10,000+ monthly transactions with 99% accuracy**.
- Leveraged **SQL and Python** to extract, transform, and analyze operational data, identifying key efficiency drivers that led to a **35% improvement in delivery speed** and **25% increase in adoption**.
- Collaborated with **Product, Infrastructure, and Security** teams to establish **data collection standards and reporting frameworks** for continuous monitoring and optimization.
- Implemented **robustness-focused engineering practices** (automated testing, debugging protocols, adversarial input handling, continuous monitoring) and applied a **product-led mindset** through **20+ stakeholder interviews**, improving system reliability and accelerating feature delivery.

Software Engineer | Art Sphere | Dec 2023 - Feb 2024 (Contract)

- Developed a full-stack employee management system with a dynamic front-end UI to support over 100 staff members, automating time-tracking and project info access, **saving the company $20K+ annually and increasing user adoption by 10% and engagement by 25%.**

Cyber Security Analyst | BBS | Istanbul, Turkey | July - Aug 2022

- Enhanced system resilience by executing penetration tests and DDoS simulations, leading to a **25% faster incident response** and preventing **thousands of potential malicious requests daily** through firewall optimization.

PROJECTS & RESEARCH

Who Congress Actually Listens to on Artificial Intelligence - Built a data-driven policy analysis project examining 300+ Congressional and federal AI documents to identify which actors most influence AI policymaking. Combined semantic retrieval, named-entity recognition, and a custom influence model to quantify institutional and individual impact. Findings showed federal agencies accounted for ~80% of total influence, challenging dominant media narratives**. Translated analysis into public-facing journalism** using **Datawrapper and Flourish** to produce interactive charts and visual explainers for broad audiences. Who Congress Actually Listens to on Artificial Intelligence

Media Bias & Public Perception of Artificial Intelligence - Analyzed 10,000+ global news headlines using Python, TextBlob, and scikit-learn to model sentiment and topics in AI coverage. Identified bias patterns across outlet ideology and editorial tone, examining how media framing and machine interpretation impact public trust and misinformation risk.

Spam Detection Model - Built and optimized ML models (Naive Bayes, Random Forest, KNN) for spam detection in large-scale Twitter datasets, focusing on improving model robustness and minimizing false positives under adversarial data conditions. **Achieved F1 score of .9 / 90% Accuracy** - Spam Tweets.ipynb.